Caloric Requirements as Registry Bit-Rate Maintenance—Deriving 3000 kcal from 6-Bit Existence Twist and 512-Bit Operational Overhead

Proving Metabolism = Manifold Coherence Tax Not Chemical Combustion, Food = Signal Integrity Fuel for Topological Stability

Registry: [CKS-BIO-1-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-13-2026] → [CKS-MATH-16-2026] → [CKS-DWDM-5-2026] → [CKS-MATH-17-2026] → [CKS-MATH-18-2026] → [CKS-MATH-19-2026] → [CKS-MATH-20-2026] → [CKS-MATH-21-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Abstract

We derive caloric requirements as registry bit-rate maintenance proving metabolism = manifold coherence tax rather than chemical combustion. From substrate energy conversion, we establish: (1) 1 bit maintenance = 342 kcal/day (base energy quantum for human-scale 144-LU mesh against 2 π background, impedance cost of 84-bit word in 3-dipole gearbox), (2) 6-bit existence kernel = 2052 kcal baseline (mandatory twist preventing soliton evaporation, RAID-1 signature minimum, registry BMR before any work), (3) 90kg/180cm aperture scaling = 1.12 $\times$ multiplier (volumetric address range adjustment, Jacobian stretch factor, 5.625 $\times$ 32cm height quantization), (4) Idle power = 2298 kcal/day (6 bits $\times$ 342 $\times$ 1.12, substrate maintenance for 90kg mesh, below this = brownout mode), (5) 512-bit sovereign overhead = +684 kcal (2 additional bits for admin

access, active write capability, high-bandwidth processing load), (6) Total operational = 2982 kcal/day (8.72 bits total, maintains zero-remainder state, enables line-rate processing, prevents parity drop), (7) Starvation < 2050 kcal = registry de-allocation (6-bit twist frays, SNR collapses, consciousness down-clocks to 66-bit survival, LERP→saccade reversion), (8) Hunger = voltage brownout (not chemical depletion but signal power drop, processor throttling, admin privileges suspended to save kernel), (9) "Hangry" = 66-bit decoherence (system prioritizes existence over bandwidth, emotional regulation disabled, coherence sacrificed for survival), (10) Food = voltage rail fuel (maintains transceiver power, sustains impedance matching, prevents topological decay, enables zero-latency execution). Metabolism proven as topological maintenance not combustion—eating = powering substrate signature, 3000 kcal = operational requirement for 512-bit sovereign function.

Key Result: Calories = bit voltage | 6 bits = existence | 2 bits = sovereign | 3000 kcal = operational | Below = brownout | Food = signal power

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Part I: The Energy-Bit Conversion

1. The Fundamental Quantum

From substrate mechanics:

Why energy required:

Physical soliton exists:

Against 2 π background

Requires constant refresh

1/32 Hz cycle

Without energy input:

Topology decays

Bits unravel

Soliton evaporates

Death occurs

The bit-rate tax:

Each bit maintained:

Costs energy per cycle

To prevent decoherence

Against entropy

Universal constant:

For human-scale mesh

In 1.0G field

At 15.19ms render rate

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2. The 342 kcal/bit Derivation

Base energy quantum:

From 144-LU mesh:

Standard human surface:

144 Logos Units

12^2 node structure

Bilateral manifold

Impedance cost:

84-bit word

Through 3-dipole gearbox

D=3 coordination

Heat generated:

Information friction

Must be compensated

Per bit per day

The calculation:

Base thermal minimum:

28.5 kcal/bit baseline

12-bond coherence multiplier:

From electron loop

12-node stability

$\times$ 12 factor

Result:

$28.5 \times 12 = 342$ kcal/bit/day

This is:

Energy to maintain

One bit of twist

In 3D manifold

Against substrate

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3. Why Not Chemical Combustion

Classical metabolism wrong:

Combustion model:

Claims:

Food = fuel

Oxidized for energy

ATP production

Chemical bonds

Problems:

Why exact amounts?

Why individual variation?

Why mental state affects?

Why purpose matters?

Topological model:

Reality:

Food = energy substrate

Maintains bit signature

Prevents decay

Registry refresh

Explains:

Exact quantum requirements

Coherence dependency

Mental state coupling

Purpose integration

Part II: The 6-Bit Existence Cost

4. The Mandatory Kernel

What the 6 bits are:

Derived from previous work:

Minimum twist:

To anchor into 3D

From 2D substrate

Creates depth (z-axis)

RAID-1 signature:

Bilateral verification

S=2 parity check

Both sides must match

Without 6 bits:

Signature fails

Soliton evaporates

Registry rejects

Death immediate

The kernel tax:

Not optional:

Must pay per cycle

1/32 Hz rate

Every 31.25ms

Even when idle:

Sleeping

Coma

Meditation

Still costs 6 bits

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5. Baseline BMR Calculation

Base metabolic rate:

The formula:

BMR = Bit_cost × Energy_quantum

Where:

Bit_cost = 6 (existence kernel)

Energy_quantum = 342 kcal/bit

Calculation:

6 × 342 = 2052 kcal/day

This is BMR:

Standard medicine measures:

~2000 kcal for average male

"Unexplained" why this number

CKS derives:

Exactly from 6-bit requirement

342 kcal quantum

2052 kcal result

Perfect match:

Not coincidence

Mathematical necessity

Substrate requirement

Part III: Aperture Scaling

6. The 90kg/180cm Multiplier

Volume adjustment:

Why scaling needed:

Different body sizes:

More mass = more nodes

More height = more stretch

Requires adjustment

90kg/180cm specific:

Larger than average

More topological volume

Higher maintenance cost

The Jacobian factor:

From J ≈ 7.7:

Height quantization

180cm = 5.625 × 32cm

(32 = word length)

Volume calculation:

Mass/height ratio

Relative to ideal

Result:

M_scale = 1.12

12% increase

For 90kg/180cm

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7. Scaled Idle Power

Adjusted baseline:

The calculation:

Scaled BMR = Base × M_scale

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= 2052  $\times$  1.12

= 2298 kcal/day
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What this represents:

Substrate idle power:

For 90kg mesh

180cm aperture

Doing zero work

Just existing

Below this threshold:

Brownout begins

System throttles

Coherence drops

Hunger emerges

Part IV: The Sovereign Overhead

8. Active Processing Load

512-bit operation cost:

Why more required:

Standard 84-bit:

Basic function:

Read-only access

Passive sampling

Minimal processing

2298 kcal sufficient

Active 512-bit:

Admin operations:

Writing papers

Active defense

High-bandwidth cognition

Physical readiness

Adds bit density:

More data through antenna

Registry friction generated

Must flush continuously

The overhead:

Additional bits needed:

2 bits for active signature

Sovereign access level

Admin write capability

Energy cost:

2 × 342 = 684 kcal/day

Total bits:

6 (existence) + 2 (sovereign)

= 8.72 bits total

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9. Total Operational Power

Complete requirement:

The final calculation:

Total = Idle + Active

= 2298 + 684

= 2982 kcal/day

Rounded operational:

~3000 kcal/day

For:

90kg/180cm frame

512-bit operation

Zero-latency readiness

Continuous admin access

Why this exact amount:

Not arbitrary:

8.72 bits × 342 kcal

× 1.12 scale factor

Mathematical necessity

Maintains:

Zero-remainder state (R=0)

Full bandwidth

Line-rate processing

Parity integrity

Part V: Starvation Mechanics

10. The Brownout Threshold

What happens below 2298:

Progressive degradation:

2298-2000 kcal (mild deficit):

System response:

Throttle non-essential

Reduce 512-bit processing

Maintain 84-bit core

Observable:

Mental fatigue

Reduced creativity

Slower thinking

Irritability begins

2000-1800 kcal (moderate deficit):

System response:

Further throttling

Admin access suspended

Focus on survival

Observable:

“Hangry” state

Emotional dysregulation

SNR collapse

Coherence drops

Below 1800 kcal (severe deficit):

System response:

Emergency mode

66-bit decoherence

Survival priority only

Observable:

Cannot focus

Anxiety/panic

Physical weakness

Personality change

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11. Registry De-allocation

Below 2050 kcal:

The 6-bit fraying:

Cannot maintain:

Full existence kernel

Bits begin dropping

Topology weakening

Physical effects:

Muscle catabolism

(Sacrificing nodes)

Organ throttling

(Reducing address space)

Brain fog

(Processor down-clock)

Why 2050 specific:

6 bits $\times$ 342 = 2052:

Below this

Cannot pay existence tax

Registry begins eviction

Starvation = registry de-allocation:

System reclaims nodes

To maintain core

Death when core fails

Part VI: Hunger Mechanics

12. Hunger as Voltage Alert

Not chemical signal:

Classical explanation:

Low blood sugar:

Triggers hunger

Chemical receptor

Homeostatic response

Problem:

Why variable timing?

Why mental state affects?

Why purpose overrides?

CKS explanation:

Voltage brownout detection:

Registry monitors power

Predicts shortage

Issues alert

Hunger = warning:

“Voltage dropping”

“Bits at risk”

“Refuel needed”

Not chemical:

But topological

Coherence monitoring

Predictive system

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13. Why Purpose Overrides Hunger

The coherence factor:

Strong purpose:

High coherence state:

Clear identity

Stable f_post

Low noise

More efficient:

Less wasted energy

Lower overhead

Better impedance match

Can operate longer:

On same calories

Via efficiency

Reduces brownout risk

No purpose (searching):

Identity jitter:

Unstable f_post

High noise

Constant searching

Very inefficient:

Energy wasted

High overhead

Poor impedance

Hungry sooner:

Same calories

More waste

Earlier brownout

Part VII: The 512-Bit Requirement

14. Why 3000 Not 2000

Standard recommendations wrong:

2000 kcal standard:

For average person:

84-bit operation

Passive existence

No admin access

Survival mode

Sufficient for:

Basic function

Sedentary life

No high-bandwidth work

3000 kcal sovereign:

For 512-bit walker:

Active admin

Writing capability

Physical defense

Zero-latency ready

Required for:

Boat jump calculus

120 papers/year

Continuous coherence

Line-rate processing

The difference:

1000 kcal = ~3 bits

These 3 bits:

Enable admin access

Maintain zero-R state

Prevent parity drop

Sustain bandwidth

Without them:

Revert to 84-bit

Lose LERP capability

Saccadic hunting only

Sovereign access denied

15. Eating as Voltage Maintenance

Not pleasure but necessity:

The understanding:

Food = signal voltage:

Powers transceiver

Maintains rails

Prevents brownout

3000 kcal = requirement:

For 512-bit operation

Not optional

Not negotiable

Mathematical necessity

Operational protocol:

Morning:

Restore from sleep debt

High-protein anchor

Voltage stabilization

Midday:

Maintain under load

Continuous fueling

Prevent dip

Evening:

Replenish reserves

Prepare for recovery

Reset for next cycle

Part VIII: Practical Application

16. The Paternal Feeding Protocol

For zero-latency readiness:

Not eating for pleasure:

Purpose:

Maintain transceiver power

Enable instant response

Protect family

Discipline:

Regular intervals

Sufficient quantity

Quality substrates

No brownouts

The 10kg recovery calculus:

Requires:

Full 512-bit bandwidth

Zero processing lag

Instant physical response

Cannot achieve:

On 2000 kcal

In brownout state

With throttled processor

Must maintain:

3000 kcal daily

Voltage rails stable

Admin access ready

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17. Quality vs Quantity

Substrate composition matters:

Not all calories equal:

High-quality (protein/fat):

Clean energy conversion

Low processing overhead

Stable voltage

Long burn time

Low-quality (processed carbs):

Noisy conversion

High overhead

Voltage spikes

Quick crash

Optimal mix:

Protein: 30-40%

Structural maintenance

Node repair

Mesh integrity

Fat: 30-40%

Dense energy

Clean burn

Stable voltage

Carbs: 20-40%

Quick access

Buffer fill

Activity fuel

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18. Fasting Considerations

Strategic voltage management:

Short fasts (16-24h):

Can work if:

High baseline coherence

Low noise operation

Efficient systems

Benefits:

Autophagy activation

System cleanup

Noise reduction

Risks:

Brownout if extended

Performance drop

Coherence loss

Extended fasts (>24h):

Dangerous for 512-bit:

Cannot maintain overhead

Admin access lost

Revert to survival

Only if:

Deliberately throttling

Seeking 84-bit baseline

Not operationally critical

Not compatible:

With paternal readiness

Zero-latency requirements

Sovereign function

Part IX: Verification

19. Observable Correlations

Testable predictions:

Prediction 1: Performance vs calories:

Track:

Daily caloric intake

Cognitive performance

Physical capability

Emotional regulation

Expected:

Linear improvement 2000→3000

Plateau at optimal

Decline above waste threshold

Test:

Measure response times

Problem-solving speed

Coherence metrics

Prediction 2: Hunger timing:

High coherence individuals:

Longer between hunger signals

More efficient operation

Lower overhead

Low coherence individuals:

Frequent hunger

Inefficient processing

High waste

Test:

Measure identity stability

Correlate with hunger frequency

Control for activity level

20. The Metabolic Table

Complete energy budget:

State	Bit Cost	kcal/day	Function
Death	0 bits	0	No signature
Coma	3-4 bits	1200-1500	Minimal kernel
Survival	6 bits	2050	Existence only
Idle 90kg	6.72 bits	2298	Baseline maintenance
Standard active	7-7.5 bits	2400-2600	Normal activity
Sovereign	8.72 bits	2982	512-bit operation
Overload	>9 bits	>3200	Waste/inefficiency

Conclusion

Caloric requirements derived as registry bit-rate maintenance proving metabolism = manifold coherence tax rather than chemical combustion. From substrate energy conversion: 1 bit maintenance = 342 kcal/day (base energy quantum for 144-LU mesh against 2 π background, impedance cost of 84-bit word in 3-dipole gearbox). 6-bit existence kernel = 2052 kcal baseline (mandatory twist preventing soliton evaporation, RAID-1 signature minimum, registry BMR). 90kg/180cm aperture = 1.12 $\times$ multiplier (volumetric address scaling, Jacobian stretch, height quantization at 5.625 $\times$ 32cm). Idle power = 2298 kcal/day (6 bits $\times$ 342 $\times$ 1.12, substrate maintenance before work). 512-bit sovereign overhead = +684 kcal (2 additional bits for admin access, active write capability, high-bandwidth load). Total operational = 2982 kcal/day (8.72 bits, maintains R=0 state, enables line-rate processing, prevents parity drop). Starvation < 2050 = registry de-allocation (6-bit twist frays, SNR collapses, consciousness throttles to 66-bit, LERP→saccade reversion). Hunger = voltage brownout warning (not chemical but topological, coherence monitoring, predictive alert). "Hangry" = 66-bit decoherence (system prioritizes survival over bandwidth, admin suspended, emotional regulation disabled). Food = voltage rail fuel (maintains transceiver power, prevents topological decay, enables zero-latency execution). Purpose improves efficiency (high coherence = lower overhead, stable f_post = better impedance, can operate longer on same input). 3000 kcal proven as operational requirement for 90kg/180cm sovereign maintaining continuous admin access and zero-latency readiness. Metabolism = topological maintenance not combustion, eating = powering substrate signature, hunger = voltage management.

Q.E.D.

Calories = bit voltage

6 bits = existence

2 bits = sovereign overhead

342 kcal = 1 bit quantum

2298 = idle 90kg

2982 = active sovereign

3000 = operational target

Below = brownout

Food = signal power

Purpose = efficiency

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